

Differential profile of the continuous extension of the logarithmic gap

Supplementary analytical note
to the conditional model version 16.4

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Status of this document

This text is a supplementary analytical note. It is not part of the formal proof chain of version 16.4 and is not proposed as an independent proof of Fermat's Last Theorem.

The purpose of the note is to investigate the first and second derivatives of the continuous extension of the logarithmic gap, which was already defined in the conditional algebraic-logarithmic model.

Abstract

This note investigates the differential behavior of the continuous extension of the sequence of logarithmic gaps

$$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right).$$

After replacing the discrete exponent n with a real variable $x > 0$, the following function is introduced:

$$\Delta(x) = \frac{1}{x} \ln \left(\frac{2^{x-1}}{x} \right) = \ln 2 - \frac{\ln(2x)}{x}.$$

Its first and second derivatives are strictly calculated. It is proved that on the domain $x \geq 2$ the function is positive after the point $x = 2$, strictly increases, and approaches the horizontal asymptote $\ln 2$. The inflection point has the exact value

$$x_0 = \frac{e^{3/2}}{2}.$$

After this point, the function becomes concave, and for all integer expo-

nents $n \geq 3$, the following holds:

$$\Delta''(n) < 0.$$

The negativity of the second derivative does not mean the negativity of the logarithmic gap itself. It merely means that the positive and increasing function continues to grow at a decreasing rate. Additionally, the ratio of two residual scales is investigated, defined by the formula:

$$R(x) = \exp(\Delta(x)).$$

The obtained results clarify the analytical structure of the model, but they do not prove that a hypothetical positive integer solution to Fermat's equation must satisfy the condition of a zero logarithmic gap.

Keywords: logarithmic gap; continuous extension; first derivative; second derivative; inflection point; monotonicity; concavity; horizontal asymptote; residual factors; ratio of residual scales; binomial symmetry; zero logarithmic gap; asymptotic analysis; conditional model of Fermat's Last Theorem; physical mathematics.

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1 Initial discrete quantity

In the conditional model of version 16.4, comparing two real normalizations of a single odd binomial kernel leads to the exact relation

$$l^n = q^n \frac{2^{n-1}}{n},$$

where $l > 0$ and $q > 0$ are two positive real residual factors.

Since both factors are positive, the logarithmic gap is defined:

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right). \quad (1.1)$$

For positive integer exponents, the following holds:

$$\Delta_1 = 0, \quad \Delta_2 = 0,$$

and for every integer $n > 2$

$$\Delta_n > 0.$$

The condition of a zero gap is equivalent to

$$\Delta_n = 0 \quad \Longleftrightarrow \quad \frac{2^{n-1}}{n} = 1 \quad \Longleftrightarrow \quad 2^{n-1} = n.$$

In this note, this condition is not re-justified. Another problem is investigated: how the quantity Δ_n behaves if the exponent is considered as a real variable.

2 Transition from a discrete exponent to a real variable

The derivative is defined for a function of a real variable. Therefore, one should not speak of differentiating the integer exponent n without a prior extension of the original sequence.

Let us introduce the function

$$\Delta(x) := \frac{1}{x} \ln \left(\frac{2^{x-1}}{x} \right), \quad x > 0. \quad (2.1)$$

For each positive integer n , it coincides with the original sequence:

$$\Delta(n) = \Delta_n.$$

Using the properties of the logarithm, let us transform (2.1):

$$\begin{aligned}
\Delta(x) &= \frac{1}{x} [(x-1) \ln 2 - \ln x] \\
&= \ln 2 - \frac{\ln 2 + \ln x}{x} \\
&= \ln 2 - \frac{\ln(2x)}{x}.
\end{aligned} \tag{2.2}$$

Form (2.2) is particularly convenient for differentiation.

3 General shape of the graph on the domain

$$x > 0$$

Before restricting ourselves to the domain $x \geq 2$, it is useful to briefly consider the behavior of the function over the entire positive semi-axis.

From

$$\Delta(x) = \frac{1}{x} \ln \left(\frac{2^{x-1}}{x} \right)$$

it can be seen that the zeros of the function are determined by the equation

$$2^{x-1} = x.$$

The positive real solutions to this equation are

$$x = 1 \quad \text{and} \quad x = 2.$$

On the interval $1 < x < 2$ it holds that

$$\Delta(x) < 0,$$

whereas on the domain $x > 2$

$$\Delta(x) > 0.$$

For the conditional model version 16.4, the domain of primary interest is

$$x \geq 2,$$

since the point $x = 2$ corresponds to the last zero value of the gap, and all integer exponents $n > 2$ lie to the right of it.

4 First derivative

Let us differentiate the function

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}.$$

The derivative of the expression

$$\frac{\ln(2x)}{x}$$

is equal to

$$\frac{\frac{1}{x} \cdot x - \ln(2x)}{x^2} = \frac{1 - \ln(2x)}{x^2}.$$

Consequently,

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}. \quad (4.1)$$

4.1 Sign of the first derivative

The denominator x^2 is positive for $x > 0$. Therefore, the sign of the first derivative is determined by the numerator:

$$\ln(2x) - 1.$$

The equality

$$\Delta'(x) = 0$$

is equivalent to

$$\ln(2x) = 1,$$

whence

$$x = \frac{e}{2} \approx 1.35914. \quad (4.2)$$

Consequently,

$$\Delta'(x) < 0 \quad \text{for } 0 < x < \frac{e}{2},$$

and

$$\Delta'(x) > 0 \quad \text{for } x > \frac{e}{2}.$$

In particular, for the entire domain of interest $x \geq 2$,

$$\ln(2x) \geq \ln 4 > 1,$$

therefore

$$\Delta'(x) > 0 \quad (x \geq 2). \quad (4.3)$$

Theorem 4.1 (Strict monotonicity after the quadratic case). *The function*

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}$$

strictly increases on the interval

$$[2, +\infty).$$

Proof. For each $x \geq 2$

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}.$$

Since

$$\ln(2x) \geq \ln 4 > 1,$$

the numerator is positive. The denominator is also positive. Therefore

$$\Delta'(x) > 0.$$

Consequently, $\Delta(x)$ strictly increases. □

Since

$$\Delta(2) = 0,$$

strict monotonicity implies

$$\Delta(x) > 0 \quad (x > 2). \tag{4.4}$$

For integer exponents we obtain

$$0 = \Delta_2 < \Delta_3 < \Delta_4 < \dots .$$

4.2 Limit of the first derivative

From (4.1) it follows that

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}.$$

The logarithm grows slower than any positive power of x , therefore

$$\lim_{x \rightarrow \infty} \Delta'(x) = 0. \tag{4.5}$$

This means that the function continues to increase, but its growth rate gradually decreases and tends to zero.

It is important to note that the existence of a finite limit of the function

does not automatically follow from the mere equality

$$\Delta'(x) \rightarrow 0.$$

For example, the derivative of the function $\ln x$ also tends to zero, although the function itself increases unboundedly.

The finite limit of $\Delta(x)$ is established directly from the exact formula

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}.$$

5 Second derivative

Let us differentiate

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}.$$

Let

$$u(x) = \ln(2x) - 1, \quad v(x) = x^{-2}.$$

Then

$$u'(x) = \frac{1}{x}, \quad v'(x) = -2x^{-3}.$$

Consequently,

$$\begin{aligned} \Delta''(x) &= \frac{1}{x}x^{-2} - 2(\ln(2x) - 1)x^{-3} \\ &= \frac{1 - 2\ln(2x) + 2}{x^3} \\ &= \frac{3 - 2\ln(2x)}{x^3}. \end{aligned} \tag{5.1}$$

5.1 Inflection point

The condition

$$\Delta''(x) = 0$$

is equivalent to

$$3 - 2\ln(2x) = 0.$$

Hence

$$\ln(2x) = \frac{3}{2},$$

therefore

$$x_0 = \frac{e^{3/2}}{2} \approx 2.24084. \tag{5.2}$$

Thus,

$$\Delta''(x) > 0 \quad \text{for } 0 < x < \frac{e^{3/2}}{2},$$

and

$$\Delta''(x) < 0 \quad \text{for } x > \frac{e^{3/2}}{2}.$$

On the domain of interest:

$$\Delta''(x) > 0 \quad \text{for } 2 \leq x < \frac{e^{3/2}}{2},$$

$$\Delta''(x) = 0 \quad \text{for } x = \frac{e^{3/2}}{2},$$

and

$$\Delta''(x) < 0 \quad \text{for } x > \frac{e^{3/2}}{2}.$$

Since

$$2 < \frac{e^{3/2}}{2} < 3,$$

for each integer exponent $n \geq 3$ the following holds:

$$\Delta''(n) < 0. \tag{5.3}$$

Theorem 5.1 (Concavity for exponents of three or higher). *The continuous extension $\Delta(x)$ is strictly concave on the interval*

$$[3, +\infty).$$

Proof. For $x \geq 3$,

$$2x \geq 6.$$

Consequently,

$$\ln(2x) \geq \ln 6 > \frac{3}{2}.$$

Therefore

$$3 - 2\ln(2x) < 0.$$

Since $x^3 > 0$, from (5.1) it follows that

$$\Delta''(x) < 0.$$

□

5.2 Limit of the second derivative

From

$$\Delta''(x) = \frac{3 - 2\ln(2x)}{x^3}$$

it follows that

$$\lim_{x \rightarrow \infty} \Delta''(x) = 0. \quad (5.4)$$

For large x , the second derivative approaches zero from the negative side:

$$\Delta''(x) \rightarrow 0^-.$$

This means that the graph remains concave, but its curvature gradually weakens.

6 Why the negative second derivative does not create a contradiction

Three different properties of the function should be distinguished:

1. the sign of the function itself $\Delta(x)$;
2. the sign of the first derivative $\Delta'(x)$;
3. the sign of the second derivative $\Delta''(x)$.

For $x > 2$, the following simultaneously hold:

$$\Delta(x) > 0$$

and

$$\Delta'(x) > 0.$$

Starting from the point

$$x > \frac{e^{3/2}}{2},$$

the following also holds:

$$\Delta''(x) < 0.$$

These statements do not contradict each other.

The condition

$$\Delta(x) > 0$$

means that the function value is positive.

The condition

$$\Delta'(x) > 0$$

means that the function is increasing.

The condition

$$\Delta''(x) < 0$$

means that the growth rate is decreasing.

Thus, the function remains positive and increasing, but each subsequent increment becomes smaller than the previous one.

Simple example

The function

$$f(x) = \ln x$$

for $x > 1$ is positive:

$$\ln x > 0.$$

Its first derivative is also positive:

$$f'(x) = \frac{1}{x} > 0.$$

However, the second derivative is negative:

$$f''(x) = -\frac{1}{x^2} < 0.$$

Consequently, the positivity of a function and the negativity of its second derivative are perfectly compatible.

In addition, the logarithm itself can take negative values:

$$\ln x < 0 \quad \text{for } 0 < x < 1.$$

It is zero at

$$x = 1$$

and positive for

$$x > 1.$$

7 Horizontal asymptote

From the exact formula

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}$$

and the standard limit

$$\lim_{x \rightarrow \infty} \frac{\ln(2x)}{x} = 0$$

we obtain

$$\lim_{x \rightarrow \infty} \Delta(x) = \ln 2. \quad (7.1)$$

Consequently, the line

$$y = \ln 2 \quad (7.2)$$

is a horizontal asymptote of the graph.

At the same time,

$$\ln 2 - \Delta(x) = \frac{\ln(2x)}{x} > 0. \quad (7.3)$$

Thus, the function approaches the asymptote from below.

The complete asymptotic picture as $x \rightarrow \infty$ has the form

$$\Delta(x) \rightarrow \ln 2,$$

$$\Delta'(x) \rightarrow 0^+,$$

$$\Delta''(x) \rightarrow 0^-.$$

8 Rate of approach to the limit

The equality

$$\ln 2 - \Delta(x) = \frac{\ln(2x)}{x}$$

not only establishes the limit, but also exactly sets the rate of approach to it.

For large x ,

$$\ln(2x) = \ln x + \ln 2,$$

therefore

$$\ln 2 - \Delta(x) \sim \frac{\ln x}{x}.$$

The first derivative has the asymptotics

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2} \sim \frac{\ln x}{x^2}.$$

The second derivative has the asymptotics

$$\Delta''(x) = \frac{3 - 2\ln(2x)}{x^3} \sim -\frac{2\ln x}{x^3}.$$

Consequently:

- the distance to the asymptote decreases approximately as $\ln x/x$;
- the growth rate of the function decreases approximately as $\ln x/x^2$;
- the curvature decreases in absolute value approximately as $2\ln x/x^3$.

9 Ratio of residual scales

From the definition of the logarithmic gap

$$\Delta(x) = \ln \frac{l}{q}$$

the ratio formally follows:

$$\frac{l}{q} = \exp(\Delta(x)).$$

Let us introduce the function

$$R(x) := \exp(\Delta(x)). \quad (9.1)$$

Using the formula for $\Delta(x)$, we obtain

$$\begin{aligned} R(x) &= \exp \left[\frac{1}{x} \ln \left(\frac{2^{x-1}}{x} \right) \right] \\ &= \left(\frac{2^{x-1}}{x} \right)^{1/x} \\ &= \frac{2}{(2x)^{1/x}}. \end{aligned} \quad (9.2)$$

At $x = 2$,

$$R(2) = 1.$$

For $x > 2$, since $\Delta(x) > 0$,

$$R(x) > 1.$$

The first derivative of the function $R(x)$ is

$$R'(x) = R(x)\Delta'(x). \quad (9.3)$$

Since on the domain $x \geq 2$,

$$R(x) > 0$$

and

$$\Delta'(x) > 0,$$

we obtain

$$R'(x) > 0.$$

Consequently, the ratio of residual scales strictly increases.

From

$$\Delta(x) \rightarrow \ln 2$$

it follows that

$$\lim_{x \rightarrow \infty} R(x) = e^{\ln 2} = 2. \quad (9.4)$$

Thus,

$$1 = R(2) < R(x) < 2 \quad (x > 2),$$

and

$$R(x) \rightarrow 2.$$

10 Is it possible to differentiate $l(x)$ and $q(x)$ separately?

If we formally assume that there exist positive smooth functions $l(x)$ and $q(x)$ such that

$$\Delta(x) = \ln \frac{l(x)}{q(x)},$$

then

$$\Delta'(x) = \frac{l'(x)}{l(x)} - \frac{q'(x)}{q(x)}. \quad (10.1)$$

The second derivative would have the form

$$\Delta''(x) = \frac{l''(x)}{l(x)} - \left(\frac{l'(x)}{l(x)} \right)^2 - \frac{q''(x)}{q(x)} + \left(\frac{q'(x)}{q(x)} \right)^2. \quad (10.2)$$

However, in the original model, l and q are defined for a fixed integer exponent n and also depend on the parameters m and p . Separate canonical smooth extensions

$$l(x) \quad \text{and} \quad q(x)$$

have not yet been specified.

Therefore, the equalities (10.1) and (10.2) should not be used as independent new statements of the model.

Only the ratio is strictly defined:

$$\frac{l}{q} = R(x) = \exp(\Delta(x)).$$

One can choose an arbitrary positive smooth function $q(x)$ and set

$$l(x) = q(x)R(x).$$

Then the required ratio will be satisfied, but such a choice is not unique.

Consequently, the differential analysis of the ratio l/q is unambiguous, while the differential analysis of l and q separately requires additional structure.

11 Discrete consequences

Since $\Delta(x)$ strictly increases on $[2, +\infty)$, for integers $n \geq 2$ the following holds:

$$\Delta_{n+1} > \Delta_n. \quad (11.1)$$

This can also be seen directly:

$$\begin{aligned} \Delta_{n+1} - \Delta_n &= \left[\ln 2 - \frac{\ln(2n+2)}{n+1} \right] - \left[\ln 2 - \frac{\ln(2n)}{n} \right] \\ &= \frac{\ln(2n)}{n} - \frac{\ln(2n+2)}{n+1}. \end{aligned} \quad (11.2)$$

The function

$$h(x) = \frac{\ln(2x)}{x}$$

has the derivative

$$h'(x) = \frac{1 - \ln(2x)}{x^2} < 0 \quad (x \geq 2).$$

Consequently,

$$h(n) > h(n+1),$$

which proves

$$\Delta_{n+1} - \Delta_n > 0.$$

For all integers $n \geq 3$, the following holds:

$$\Delta''(n) < 0.$$

In the continuous interpretation, this means that consecutive increments of the gap must decrease:

$$\Delta_{n+1} - \Delta_n > \Delta_{n+2} - \Delta_{n+1}$$

for exponents lying on the concave part of the function.

12 Numerical values

The table shows the values of the function and its first two derivatives at several integer points.

Table 1: Numerical differential profile of the logarithmic gap

n	Δ_n	$\Delta_n^{(2)}$	$\Delta'(n)$	$\Delta''(n)$
2	0.000000	0.000000	0.096574	0.028426
3	0.095894	0.138346	0.087973	-0.021612
4	0.173287	0.250000	0.067465	-0.018108
5	0.232630	0.335614	0.052103	-0.012841
6	0.278996	0.402506	0.041247	-0.009120
7	0.316139	0.456092	0.033450	-0.006642
8	0.346574	0.500000	0.027697	-0.004971
9	0.371995	0.536675	0.023338	-0.003814
10	0.393574	0.567807	0.019957	-0.002991

The table shows that:

- the values of Δ_n increase;
- the values of the first derivative remain positive but decrease;
- the values of the second derivative starting from $n = 3$ are negative;
- the negative values of the second derivative approach zero.

13 Graph of the logarithmic gap

Figure 1 shows the function $\Delta(x)$, its horizontal asymptote $y = \ln 2$, the quadratic point $x = 2$, and the inflection point

$$x_0 = \frac{e^{3/2}}{2}.$$

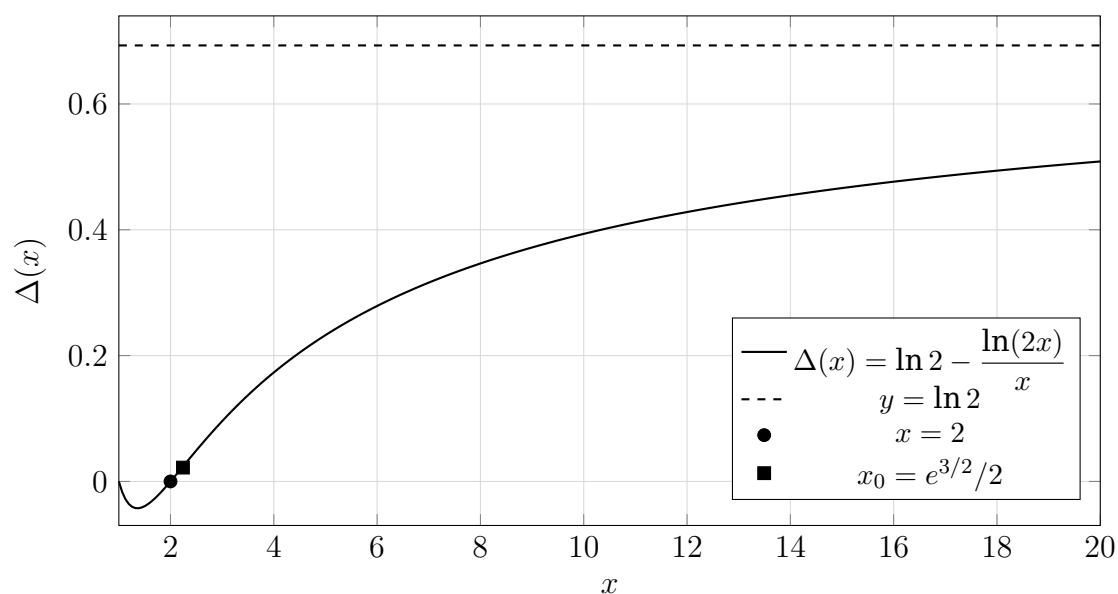


Figure 1: Continuous extension of the logarithmic gap. After the point $x = 2$, the function is positive and increasing, and after the inflection point $x_0 = e^{3/2}/2$ it becomes concave and approaches the asymptote $y = \ln 2$.

14 Graphs of the first and second derivatives

Figure 2 shows the first and second derivatives on the domain $x \geq 2$.

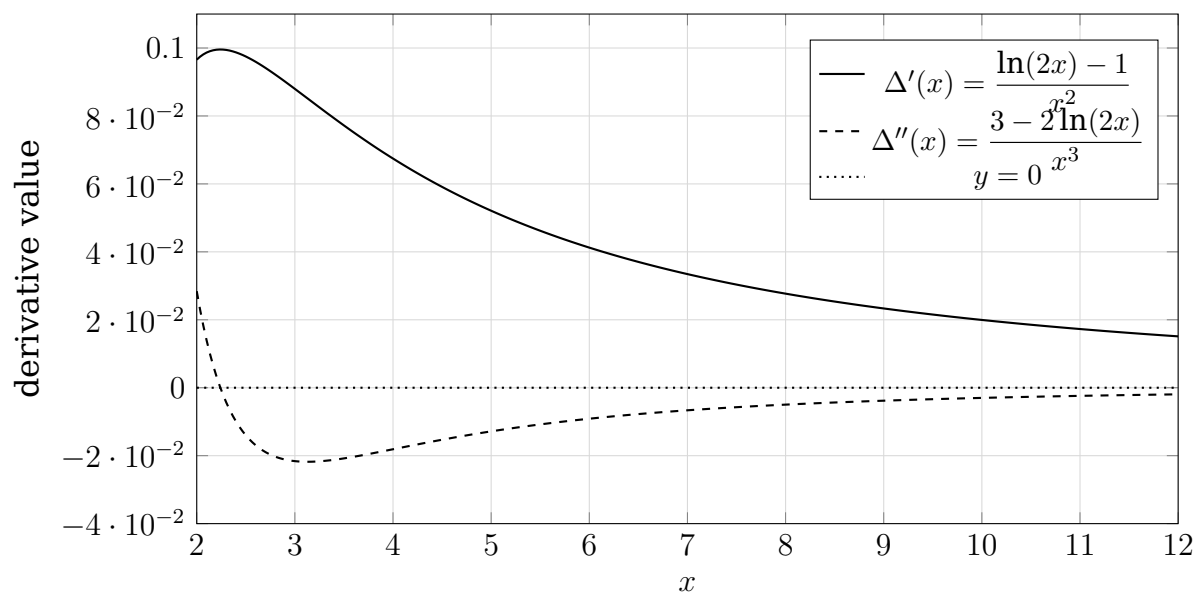


Figure 2: First and second derivatives of the logarithmic gap. The first derivative is positive and tends to zero. The second derivative changes sign between 2 and 3, after which it remains negative and also tends to zero.

15 Summary theorem

Theorem 15.1 (Differential profile of the logarithmic gap). *Let*

$$\Delta(x) = \frac{1}{x} \ln \left(\frac{2^{x-1}}{x} \right) = \ln 2 - \frac{\ln(2x)}{x}, \quad x \geq 2.$$

Then the following statements hold:

(i)

$$\Delta(2) = 0.$$

(ii) *For each* $x > 2$

$$\Delta(x) > 0.$$

(iii) *For each* $x \geq 2$

$$\Delta'(x) > 0.$$

(iv) *The function has a single inflection point on the domain* $x \geq 2$:

$$x_0 = \frac{e^{3/2}}{2}.$$

(v) *For*

$$2 \leq x < x_0$$

it holds that

$$\Delta''(x) > 0.$$

(vi) *For*

$$x > x_0$$

it holds that

$$\Delta''(x) < 0.$$

(vii) *For each integer* $n \geq 3$

$$\Delta''(n) < 0.$$

(viii)

$$\lim_{x \rightarrow \infty} \Delta(x) = \ln 2.$$

(ix)

$$\lim_{x \rightarrow \infty} \Delta'(x) = 0^+.$$

(x)

$$\lim_{x \rightarrow \infty} \Delta''(x) = 0^-.$$

(xi) *For the ratio of scales function*

$$R(x) = e^{\Delta(x)}$$

it holds that

$$R(2) = 1, \quad R'(x) > 0 \quad (x \geq 2), \quad \lim_{x \rightarrow \infty} R(x) = 2.$$

Proof. The statements follow from the exact formulas

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}$$

and

$$\Delta''(x) = \frac{3 - 2\ln(2x)}{x^3},$$

as well as from the equality

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}.$$

On the domain $x \geq 2$ we have

$$\ln(2x) \geq \ln 4 > 1,$$

whence $\Delta'(x) > 0$.

The second derivative vanishes exactly when

$$\ln(2x) = \frac{3}{2},$$

that is, when

$$x = \frac{e^{3/2}}{2}.$$

The limits of the derivatives follow from the fact that the logarithm grows slower than the power functions x^2 and x^3 . The limit of the function itself follows from

$$\frac{\ln(2x)}{x} \rightarrow 0.$$

For $R(x) = e^{\Delta(x)}$ we have

$$R'(x) = R(x)\Delta'(x) > 0$$

and

$$R(x) \rightarrow e^{\ln 2} = 2.$$

□

16 What this analysis adds to version 16.4

The differential analysis clarifies the behavior of the positive logarithmic gap for exponents higher than the quadratic one.

It is strictly established that after $x = 2$:

1. the logarithmic gap becomes positive;
2. the gap strictly increases;
3. there is a single inflection point between $x = 2$ and $x = 3$;
4. starting from the integer exponent $n = 3$, the graph is concave;
5. the growth rate of the gap gradually tends to zero;
6. the gap itself tends to a finite limit $\ln 2$;
7. the ratio of residual scales tends to 2.

This provides a more complete analytical picture of the coefficient imbalance.

However, these results do not establish the implication

$$x^n + y^n = z^n \implies \Delta_n = 0.$$

In other words, differentiation describes the behavior of an already defined function $\Delta(x)$, but does not prove that every hypothetical integer solution to Fermat's equation must be located at its zero point.

17 Logical status of the results

Table 2: What is proved and what remains conditional

Strictly proved	Does not follow from the differential analysis
Formula $\Delta'(x) = \frac{\ln(2x) - 1}{x^2}$	The necessity of the condition $\Delta_n = 0$
Formula $\Delta''(x) = \frac{3 - 2\ln(2x)}{x^3}$	for a hypothetical solution to FLT Unconditional proof of FLT
Strict monotonicity for $x \geq 2$	Integrality of the individual residual factors l and q
Inflection point $x_0 = \frac{e^{3/2}}{2}$	Uniqueness of the smooth extensions $l(x)$ and $q(x)$
Concavity for $x > \frac{e^{3/2}}{2}$	Physical or entropy status of the second derivative
Limit $\Delta(x) \rightarrow \ln 2$	Identification of the logarithmic gap with thermodynamic entropy
Limit $R(x) \rightarrow 2$	Necessity of the equality $l = q$ for any hypothetical integer solution

18 Conclusion

The continuous extension of the discrete logarithmic gap has the form

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}.$$

Its first derivative is

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2},$$

and the second derivative is

$$\Delta''(x) = \frac{3 - 2 \ln(2x)}{x^3}.$$

On the domain $x \geq 2$, the function strictly increases. At the point

$$x_0 = \frac{e^{3/2}}{2} \approx 2.24084$$

it changes the character of its curvature. For all integer exponents $n \geq 3$, the graph is concave:

$$\Delta''(n) < 0.$$

This does not mean the negativity of the logarithmic gap itself. On the contrary, for $x > 2$,

$$\Delta(x) > 0,$$

and for $x \geq 2$,

$$\Delta'(x) > 0.$$

Consequently, after the quadratic case, the function remains positive and increases, but its growth rate gradually decreases.

As $x \rightarrow \infty$,

$$\Delta(x) \rightarrow \ln 2,$$

$$\Delta'(x) \rightarrow 0^+,$$

$$\Delta''(x) \rightarrow 0^-.$$

The ratio of residual scales

$$R(x) = e^{\Delta(x)} = \left(\frac{2^{x-1}}{x} \right)^{1/x}$$

strictly increases from

$$R(2) = 1$$

to the limit

$$\lim_{x \rightarrow \infty} R(x) = 2.$$

Thus, the differential analysis shows that the positive logarithmic gap for exponents higher than the square has a regular structure: it monotonically increases, after a single inflection point it becomes concave, and approaches a finite horizontal asymptote.

This result is an independent analytical addition to the conditional model of version 16.4. It does not eliminate its main conditional point: the need to separately prove that a hypothetical positive integer solution to Fermat's equation must imply the condition

$$\Delta_n = 0.$$